



Weighted loss for imbalanced glaucoma detection: Insights from visual explanations

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ABSTRACT

Glaucoma is a leading cause of irreversible vision loss in ophthalmology, primarily resulting from damage to the optic nerve. Early detection is crucial but remains challenging due to the inherent class imbalance in glaucoma fundus image datasets. This study addresses this limitation by applying a weighted loss function to Convolutional Neural Networks (CNNs), evaluated on the standardized SMDG-19 dataset, which integrates data from 19 publicly available sources. Key performance metrics including recall, F1-score, precision, accuracy, and AUC were analyzed, and interpretability was assessed using Grad-CAM. The results demonstrate that recall increased from 60.3% to 87.3%, representing a relative improvement of 44.75%, while F1-score improved from 66.5% to 71.4% (+7.25%). Minor trade-offs were observed in precision, which declined from 74.5% to 69.6% (−6.53%), and in accuracy, which dropped from 84.2% to 80.7% (−4.10%). In contrast, AUC rose from 84.2% to 87.4%, reflecting a relative gain of 3.21%. Grad-CAM visualizations showed consistent focus on clinically relevant regions of the optic nerve head, underscoring the effectiveness of the weighted loss strategy in improving both the performance and interpretability of CNN-based glaucoma detection systems.

1. Introduction

Recent advances in the field of deep learning have been applied in various fields, including medical, to perform early detection of various medical disorders. In the field of ophthalmology, digital fundus image analysis using deep learning is gaining significant attention. One of the major medical disorders in ophthalmology is glaucoma. Glaucoma is an irreversible medical condition that occurs due to damage to the eye's optic nerve, which carries sensory information related to vision [1]. The condition is usually caused by a build-up of fluid in the front of the eye which increases the pressure inside the eye [1].

Glaucoma is often associated with intraocular pressure (IOP). Normal values range from 11 to 21 mmHg, while abnormal values that exceed 22 mmHg are considered harmful to optic nerve cells [2]. Over time, this pressure can cause damage to the vision. This vision loss starts gradually, from peripheral vision loss, then central vision, and finally total vision loss. However, glaucoma can occur even in patients with normal IOP in whom relying solely on IOP measurements is insufficient for diagnosis. This limitation highlights the need for an image-based analysis of the optic nerve head for the complete detection of glaucoma. Routine eye exams based on digital fundus images are very important for patients and can provide valuable information for a medical specialist. Characteristics of fundus images related to the appearance of glaucoma conditions can be seen through tissue loss

at the neuroretinal margin, optic nerve enlargement, unphysiological differences between the optic nerves in the two eyes, hemorrhage at the optic disc margin, thinning of the retinal nerve fiber layer, and atrophy (tissue size shrinkage) of the parapapillary tissue [3]. With the above characteristics, the deep learning model, Convolutional Neural Network (CNN), becomes very suitable for detecting glaucoma in fundus images, given its ability to extract spatial characteristics from an image. Sample fundus images are shown in Fig. 1.

However, recent studies by Hood, Heijl, and De Moraes have challenged the conventional notion that glaucomatous damage primarily begins in the peripheral visual field. Their work has shown that structural and functional deficits can also occur in the central visual field — including the macular region — even at the earliest stages of glaucoma [4,5]. This early macular involvement is clinically significant, as it directly impacts tasks such as reading and facial recognition, which are essential to daily functioning and quality of life [6].

Standard 24–2 perimetry, which samples relatively few test points in the central 10° of vision, may fail to detect these early central losses [7]. In contrast, 10–2 visual field strategies offer denser central sampling and have been shown to reveal paracentral or arcuate defects that are missed by 24–2 testing, especially in patients with mild glaucoma or suspicious discs [8,9].

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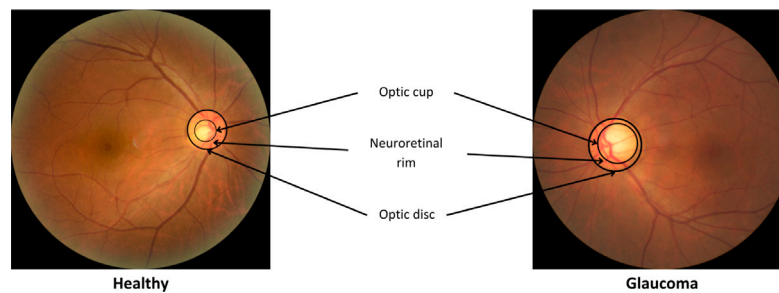


Fig. 1. Healthy and glaucoma fundus image.

In fact, studies have reported that a substantial proportion of eyes with normal 24–2 fields already show central defects on 10–2 testing [5,7]. These emphasize that glaucoma is not exclusively a peripheral condition and that central vision can be compromised before overt peripheral field loss is apparent [4,6].

In addition, CNN-based diagnosis support systems can provide relevant and important inputs that assist medical specialists in making important decisions. Therefore, various CNN architectures for early detection of eye disorders have been applied, especially those related to glaucoma [10]. This is why medical image processing has developed rapidly and become a significant tool in providing accurate results.

In general, public datasets for glaucoma detection have two main types of images, namely fundus images and OCT (Optical Coherence Tomography) images. Fundus images are usually found more in public datasets than OCT images [11]. Therefore, the performance of deep learning models will be optimized with the availability of sufficient fundus image data. This was proven by Nagaraju, Chawla, and Kumar [12] who attempted to increase the size of the dataset using data augmentation techniques and gave better results than the initial smaller dataset.

In glaucoma detection using deep learning models, unbalanced use of datasets between glaucoma and non-glaucoma images is common. Most fundus images in public datasets are healthy fundus images, while glaucoma fundus images are much less in number. An unbalanced dataset can affect model performance, especially in minority classes such as glaucoma. This was proven by Hensman and Masko [13] who showed that the distribution of training data has a large impact on CNN performance. He also found that there is a relationship between greater imbalance and poorer performance. If this problem is left unchecked, the model tends to predict well for the majority class (non-glaucoma) and under-predict the minority class (glaucoma). By addressing the problem of data imbalance, deep learning models are expected to provide better detection performance.

Class imbalance is a common issue in medical image analysis, especially for detecting conditions like glaucoma that have fewer samples. Standard loss functions often make models focus too much on the majority class, reducing sensitivity to rare diseases. Weighted loss functions help by giving more importance to underrepresented classes during training. Recent work by Umirzakova et al. [14] highlights that designing effective loss functions is essential for smart healthcare applications, particularly when data is limited or imbalanced. Our study explores the use of weighted loss in deep learning models for glaucoma detection, aiming to improve both sensitivity and overall performance. Imbalanced datasets mess up the overall feature distribution, compromise the discriminative ability of the CNN feature capabilities, and cause unacceptable errors in some cases [15]. The excellent method for handling data imbalance is class rebalancing [16] by adjusting network training by resampling [16–18].

However, the use of rebalancing and resampling in glaucoma diagnosis on unbalanced datasets poses some challenges [19]. First, this strategy has proven ineffective in accurately identifying rare cases, leading to low accuracy in practical testing, especially in early-stage glaucoma. Secondly, although the rebalancing and resampling strategy

improves classification, it unintentionally impairs the learning ability of neural network representations by distorting the original distribution, as shown by recent research in computer vision [20,21].

Weighted loss is a distinct technique from both resampling and rebalancing. It focuses on adjusting the loss function during training to handle class imbalance, providing a way to enhance model performance on underrepresented classes without altering the original dataset. Therefore, a weighted loss approach can be used to overcome the data imbalance problem. Weighted loss is a technique in deep learning model training that gives a higher weight to samples from the minority class so that they contribute equally with the majority class to the gradient calculation during model training. Research conducted by Oktavian, et al. [22] who applied the weighted loss technique to the ResNet-18 model for Alzheimer's disease detection using MRI images successfully improved the accuracy of the model to 19.2% better.

The proposed weighted loss strategy effectively addresses the class imbalance in glaucoma image datasets, yielding markedly improved sensitivity (recall) in glaucoma detection at a modest cost in precision. On the large-scale SMDG-19 dataset, the best performing CNN model trained with weighted loss achieved 44.75% higher recall and 7.25% higher F1-score compared to training the same model without weighting, albeit with an approximately 6.5% reduction in precision. Furthermore, the use of Grad-CAM provides insight into the model's decision-making by highlighting the optic disc region that most influences its predictions, thereby adding a layer of explainability to the proposed framework. In contrast to previous studies, this study uniquely evaluates the impact of a weighted loss function across multiple CNN architectures and incorporates Grad-CAM-based visual explanations, offering a more comprehensive approach that improves both glaucoma detection performance and model transparency.

Based on the previous paragraphs, this research is believed to improve the performance of the CNN model in detecting glaucoma using the weighted loss approach. The specific contributions of this study are as follows:

1. We propose the use of a weighted loss function to address the issue of class imbalance in glaucoma detection using digital fundus images.
2. We systematically evaluate the impact of this technique across three widely used CNN architectures: ResNet50, DenseNet121, and MobileNetV2.
3. We incorporate Grad-CAM visualizations to interpret the attention focus of the models, validating that the application of weighted loss leads to improved localization of clinically relevant regions, such as the optic disc and neuroretinal rim.
4. To the best of our knowledge, this is the first study to jointly assess the effect of weighted loss and model interpretability using Grad-CAM across multiple CNN backbones in the context of glaucoma classification.

2. Related works

The research conducted by Serte and Serener [23] used an image cropping approach because the optic-disc must be observed for glaucoma diagnosis. Therefore, the fundus image was cropped around the optic-disc when necessary. This is done not only for better glaucoma assessment, but also to standardize the fundus images. In addition, the dataset used by Serte and Serener was augmented so that more training data was used. This research develops a deep learning model to classify glaucoma in fundus images in general. The model was run on three different deep learning architectures, namely ResNet50, GoogLeNet, and ResNet-152. Unlike most previous studies, this research trained four different datasets simultaneously, but testing was done on each of them. The datasets used were HRF, Drishti-GS1, RIM-ONE, sjchoi86-HRF, and ACRIMA. Although each dataset showed data variability, for each of the individual datasets tested and for the three different deep learning architectures, the model showed comparable or better performance 80% of the time than previous existing research.

Research conducted by Joshi et al. [24] built a computer-aided design (CAD) system to help early identification of glaucoma. Extraction of structural characteristics from segmented optic disc and segmented optic cup can be used to characterize glaucoma and determine its severity. The main objective of the research was to estimate the potential of image analysis models for early identification and diagnosis of glaucoma. The method used in this research is an ensemble-based deep learning model for glaucoma diagnosis which is the first time developed for such a task. This research utilizes three pretrained CNN architectures for glaucoma classification. These architectures include ResNet, VGGNet, and GoogLeNet. This study utilized four distinct datasets and a combined dataset of all four to evaluate the performance of the proposed algorithm. The combined dataset was created by merging all four datasets mentioned previously. The combined dataset includes DRISHTI-GS, DRIONS-DB, HRF, and PSGIMSR. These distinct datasets were merged to form a unified dataset. Noteworthy achievements include a 91.11% accuracy on the PSGIMSR dataset, along with a sensitivity of 85.55% and specificity (probability of a negative test given that a patient is well) of 95.20% for the proposed ensemble architecture on PSGIMSR. Likewise, impressive accuracy rates of 95.63%, 98.67%, 95.64%, and 88.96% were attained for the DRIONS-DB, HRF, DRISHTI-GS, and combined dataset.

In a 2022 study by Kiefer et al. [25], advancements in automatic glaucoma detection using machine learning techniques are explored. The research comprises a literature review of recent technical articles from 2021 onward, addressing diverse image types prevalent in glaucoma datasets. The outcomes of this investigation are summarized and organized based on the most significant discoveries and methodologies. Preprocessing methods for deep learning, transfer learning, ensemble approaches, and self-organizing models are significant advances in glaucoma detection in fundus images. Transfer learning is the most common approach used by researchers due to limited resources, especially data, in building models in the medical domain. The ensemble approach improves the performance of a single classifier. One unique approach noted is the self-organizing model that can achieve performance equivalent to general deep learning models using fewer parameters and shorter training time. Kiefer recommends future research focus on standardizing algorithm performance with high-quality public datasets and exploration of multi-channel models to improve accuracy.

Research conducted by Afroze et al. [26] detected glaucoma by building a model to reduce time and cost. This research used the InceptionV3 architecture with input images processed with data augmentation techniques, such as horizontal flip and shift, shear, rotation, vertical and channel shift, and zoom. Afroze used a total of 6072 images, of which 2336 were glaucoma images and 3736 were normal (non-glaucoma) fundus images. This research obtained an accuracy of 85.29% and a value of 93.87% for AUC. However, this model tends to

detect more normal fundus images than glaucoma images due to the larger number of normal fundus images in the dataset. Afroze also conducted a comparison using DenseNet121 and ResNet50 architectures and obtained accuracies of 81.53% and 77.61%, respectively.

Research conducted by Oktavian et al. [22] addressed the challenge of classifying Alzheimer's disease using deep learning by proposing a CNN model based on the ResNet18 architecture. To overcome the limitations of a small and imbalanced dataset, the study employed transfer learning with ImageNet pre-trained weights and applied a weighted loss function to ensure balanced class representation. The researchers also experimented with replacing the standard activation function in the network with the Mish activation function to improve accuracy. The results showed that the model achieved an accuracy of 88.3 percent when using transfer learning, weighted loss, and the Mish activation function. This was a substantial improvement compared to the baseline model, which achieved only 69.1 percent accuracy. Although each component — weighted loss, transfer learning, and the Mish activation function — contributed to improved model performance individually, combining all three did not result in a significantly higher accuracy.

Other research using weighted loss functions includes notable works such as Lin et al. [27], who introduced Focal Loss, a novel loss function designed to address class imbalance in object detection tasks. Focal Loss dynamically scales the standard cross-entropy loss, reducing the relative loss for well-classified examples and focusing more on hard-to-classify examples. Buda et al. [28] compared several techniques, including weighted loss functions, to address the problem of imbalanced data. Their findings show that weighted loss functions effectively improve classification performance for unbalanced data sets, reinforcing the need for such methods in practical applications. Cui et al. [29] proposed Class-Balanced Loss which weights the loss of each class by the inverse of the effective number of samples. This method reduces the impact of class imbalance by giving higher weights to underrepresented classes. Overall, the weighted loss approach has shown significant improvement in handling class imbalance without requiring a complete overhaul of the existing technology.

3. Methods

This research begins with loading SMDG-19 fundus image data, then extracting the data, then building the model architectures. After the model is built, training and validation are conducted. Then, the models are tested to see the models' performance with standard loss function and weighted loss function. After that, a random image is visualized using Grad-CAM. Finally, the result is a classification result report using the metrics defined in Section 3.6. This research methodology is shown in Fig. 2.

3.1. Dataset

The dataset collected and used in this research is the Standardized Multi-Channel Dataset for Glaucoma (SMDG-19). SMDG-19 is a collection and standardization of 19 public datasets, consisting of fundus images along with associated image metadata, such as optic disc segmentation, optic cup segmentation, blood vessel segmentation, as well as text metadata, such as gender and age. This dataset is the largest public repository for standardized fundus images. Furthermore, not all metadata in SMDG-19 is used in this research, but only fundus image metadata related to glaucoma or healthy labels. By using images labeled 0 (healthy) or 1 (glaucoma), this dataset is ready to be processed and trained using the CNN model. The contents of SMDG-19 can be seen in Table 1.

The fundus images in SMDG-19 are standardized using a novel algorithm (Dynamic Global Foreground Threshold Algorithm) performed by Kiefer et al. by cropping the background, centering the fundus image, filling in missing information, and resizing to 512×512 pixels. This standardization ensures that most of the image information is

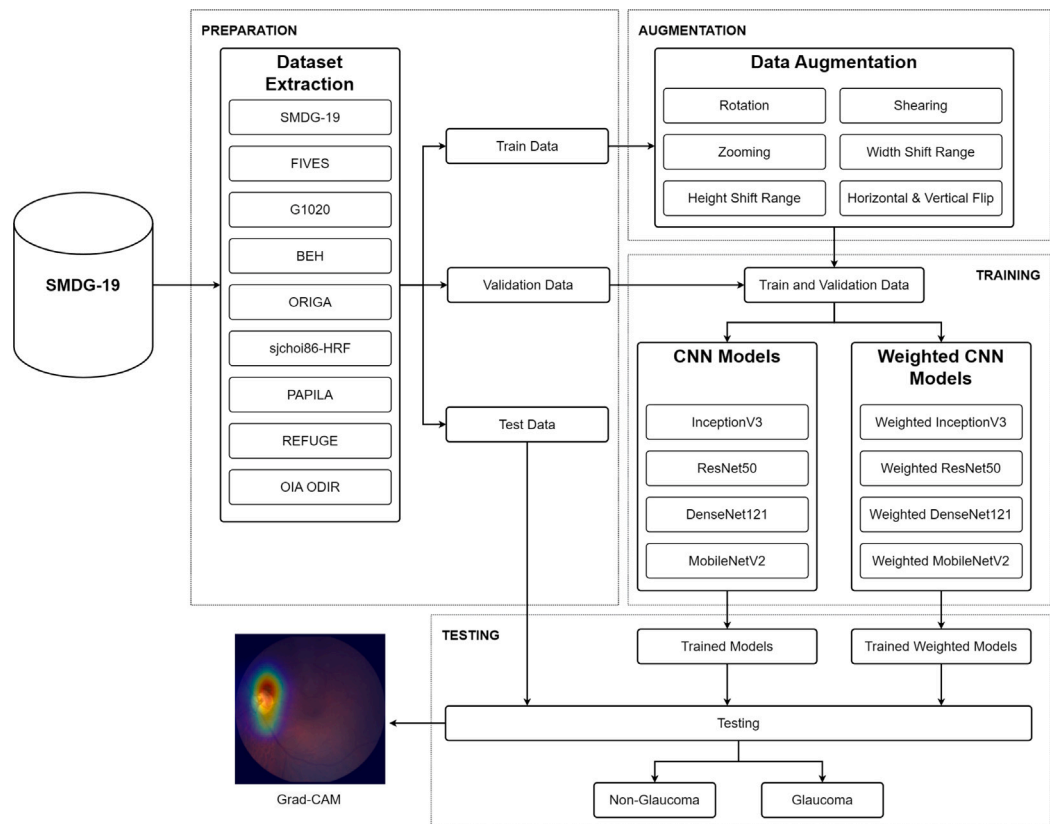


Fig. 2. Research methodology.

viable during the resizing process for machine learning-ready image processing [30].

The glaucoma and normal labels in the SMDG-19 dataset were inherited from the original annotations provided by the 19 contributing public datasets. In most instances, these labels were assigned by trained graders, ophthalmologists, or glaucoma specialists based on fundus photograph evaluation, sometimes supplemented by ancillary clinical data such as intraocular pressure or optic disc notes. However, the diagnostic criteria and labeling protocols were not standardized across datasets. As a result, the labeling process is prone to variability in interpretation and potential inconsistencies.

Diagnosing glaucoma from a single fundus image is inherently subjective and is known to suffer from limited inter-observer agreement—even among expert clinicians. Prior studies have reported that agreement among glaucoma specialists when evaluating optic nerve head appearance in fundus photographs ranges from only slight to fair, particularly in early or borderline cases. Furthermore, without confirmatory visual field testing or structural imaging (e.g., OCT), photograph-based labels may reflect uncertainty, especially in cases of physiological cupping, high myopia, or poor image quality. This heterogeneity in the labeling introduces potential label noise, which can affect model training and evaluation.

Therefore, the lack of an objective unified diagnostic standard in the 19 sources represents a limitation of this work. Although large-scale datasets like SMDG-19 enable broad generalization and training diversity, the variability in human-labeled ground truth must be acknowledged as a potential source of diagnostic ambiguity and performance ceiling in AI models. Future studies may benefit from re-annotating subsets of data using consensus labeling by panels of experts or from incorporating multi-modal ground truth that includes functional and structural confirmation to reduce subjectivity.

Table 1
SMDG-19 contents.

Dataset	Count		
	0 (non-glaucoma)	1 (glaucoma)	All
BEH	463	171	634
CRFO-v4	31	48	79
DR-HAGIS	0	10	10
DRISHTI-GS1-TRAIN	18	32	50
DRISHTI-GS1-TEST	13	38	51
EyePACS-AIROGS	0	3269	3269
FIVES	250	150	400
G1020	724	296	1020
HRF	15	15	30
JSIEC-1000	38	0	38
LES-AV	11	11	22
OIA-ODIR-TRAIN	2932	197	3129
OIA-ODIR-TEST-ONLINE	802	58	860
OIA-ODIR-TEST-OFFLINE	417	36	453
ORIGA-light	482	168	650
PAPILA	333	87	420
REFUGE1-TRAIN	360	40	400
REFUGE1-VALIDATION	360	40	400
sjchoi86-HRF	300	101	401
Total	7549	4767	12316

3.1.1. Imbalance ratio

Since SMDG-19 consists of a collection of 19 public datasets, experiments were conducted by dividing SMDG-19 based on each dataset in SMDG-19. The first part is the whole content of SMDG-19, the subsequent parts are searched from the distribution of datasets in SMDG-19. This experiment focuses on the performance of glaucoma detection as a minority label. Therefore, not all dataset in SMDG-19 are included in the experiment. This is because some dataset do not have data on any of the labels, have an equal amount of data, or have

Table 2
SMDG-19 datasets used in this research.

No.	Dataset	Count		IR
		0 (non-glaucoma)	1 (glaucoma)	
1	SMDG-19	7549	4767	1.59
2	FIVES	250	150	1.67
3	G1020	724	296	2.45
4	BEH	463	171	2.71
5	ORIGA	482	168	2.87
6	sjchoi86-HRF	300	101	2.97
7	PAPILA	333	87	3.83
8	REFUGE	720	80	9.00
9	OIA-ODIR	4151	291	14.26
Average Imbalance Ratio				4.59

a greater amount of data labeled as glaucoma. Therefore, the SMDG-19 dataset used in this research can be seen in Table 2.

The imbalance ratio (IR) is the most commonly used measure to describe the degree of imbalance in a dataset [31]. IR is defined as follows:

$$IR = \frac{N_{maj}}{N_{min}} \quad (1)$$

where N_{maj} is the sample size of the majority class and N_{min} is the sample size of the minority class. IR measures explains that when $IR = 1$, we have a perfectly balanced dataset. When $IR > 1$, the greater the IR, the greater the degree of imbalance in the dataset.

3.2. Data augmentation

Another factors that can be a problem are small amount of data and data diversity is low. This means that the model is unable to recognize an image in general due to limited training data. This is often the case with medical images which are often scarce for several reasons, such as not having many patients for some diseases, patients not allowing their images to be used, lack of medical equipment or supplies, inability to obtain images that meet the desired criteria, and inability to access images of patients living in different geographical areas or of different races [32].

Shyamalee and Meedeniya [33] used a CNN model to classify fundus images. They used several image processing and data augmentation techniques. We apply six data augmentations, adopted from their research: rotation of 10 degrees, shearing range of 0.2, zooming range of 0.1, horizontal and vertical flip, and height and width shift range of 0.1. All datasets listed in Table 2 underwent an identical data augmentation pipeline (including random rotations, flips, and brightness adjustments).

3.3. Convolutional neural network

CNN is a type of deep learning approach model that dominates in various computer vision tasks [34,35]. CNN architecture consists of several basic components in the forward propagation algorithm, such as convolution layer, pooling layer, and fully connected layer. CNNs are designed to automatically and flexibly understand the hierarchy of spatial features through backward propagation algorithms. CNNs were selected for this study due to their proven effectiveness specifically at automatically learning spatial hierarchies of features. It is particularly suited to detect subtle structural changes in the optic nerve head, such as neuroretinal rim thinning, increased cup-to-disc ratio, and localized retinal nerve fiber layer (RNFL) defects, hallmarks of glaucomatous damage in fundus photographs [36].

CNN-based architectures, including ResNet and DenseNet, have been extensively validated in ophthalmology, demonstrating high performance in tasks such as diabetic retinopathy grading, optic disc segmentation, and glaucoma screening. Their convolutional layers are effective at capturing local texture and edge-based information, which is critical for analyzing optic disc morphology and peripapillary regions.

Moreover, the availability of pre-trained weights on large-scale natural image datasets like ImageNet enables transfer learning, which allows CNNs to be efficiently fine-tuned on domain-specific medical imaging tasks, even with limited glaucoma-specific data. This reduces the need for training from scratch and improves convergence, generalization, and training efficiency. In our implementation, transfer learning helped mitigate overfitting and accelerated the optimization process.

Although emerging deep learning paradigms such as Vision Transformers (ViTs) are gaining popularity in medical imaging, CNNs remain more computationally efficient and better optimized for medium-sized datasets and low-latency inference [37]. For our task — glaucoma classification using 2D fundus photographs — CNNs offer a strong balance between performance, interpretability (e.g., via Grad-CAM visualizations), and deployment practicality.

The layers that make up the CNN architecture can be carefully constructed and designed to form an architecture that suits a variety of machine learning tasks. This work uses 4 CNN architectures, which are InceptionV3, ResNet50, DenseNet121, and MobileNetV2 with pre-trained ImageNet weights. All four architectures' last layers were fine-tuned and replaced with a sequence of layers comprised of a dropout layer of 0.5, a fully connected layer with 256 units, and a sigmoid layer.

3.4. Weighted loss function

In the standard Binary Crossentropy Loss, each sample and each class contributes equally to the overall loss calculation, which can lead to a bias towards the majority class on an unbalanced dataset. The weighted version of Binary Crossentropy Loss addresses the problem of data imbalance by giving different weights to classes based on their representation in the dataset. Classes with fewer samples are given higher weights, while classes with more samples are given lower weights so that both have an equal influence in error calculation. This approach assists the model in emphasizing attention to minority classes throughout the training phase [22]. The mathematical formula for calculating Weighted Binary Crossentropy Loss is given by:

$$J_{bce} = -\frac{1}{M} \sum_{m=1}^M w_m [y_m \ln(\hat{y}_m) + (1 - y_m) \ln(1 - \hat{y}_m)] \quad (2)$$

where M is the number of training samples, w_m is the weight of sample m , y_m is the class for sample m , and $\hat{y}_m$ is the sigmoid output. In the formula, w_m can be calculated as follows:

$$w_m = 1 - \frac{n_c}{N} \quad (3)$$

where n_c is the number of samples in class c and N is the total number of all samples from all classes.

3.5. Grad-CAM

Comprehending intricate deep learning networks can be formidable. One approach to acquiring insights into these networks involves the visualization of Class Activation Maps (CAM). CAMs help identify the specific elements in an image that the model focuses on when making predictions, shedding light on the inner workings of the model [38]. Grad-CAM [39], another technique, employs backward propagation to highlight the regions of an image that capture the model's interest. This involves utilizing the gradients of each prediction concerning the final convolutional layer of the network [40]. Because this research focuses on the effect of weighted loss, the backward propagation is adjusted so that the gradient used is the loss gradient towards the network's final convolutional layer. The Grad-CAM visualization process is depicted in Fig. 3, where w is the weight of the network's final convolutional layer and L is the discriminative saliency map.

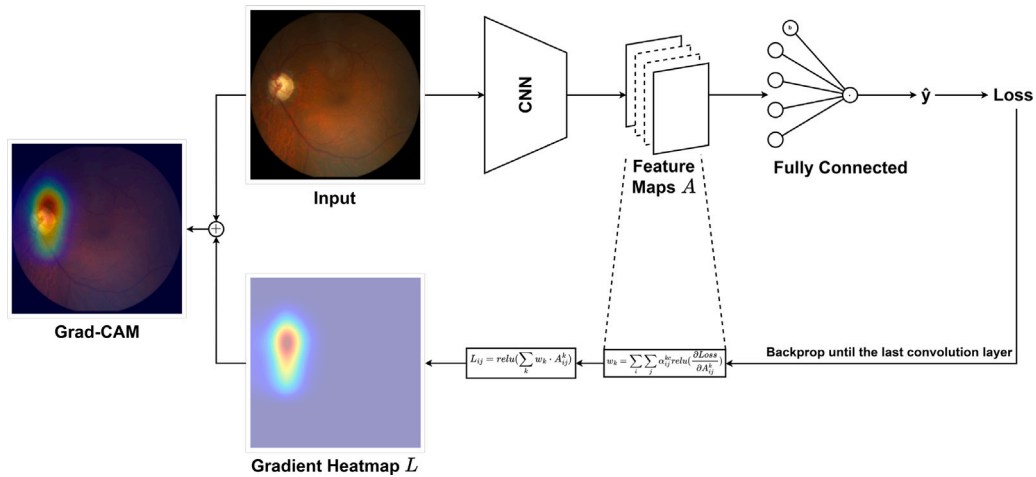


Fig. 3. Grad-CAM visualization process for glaucoma detection.

3.6. Evaluation metrics

Our experiments report precision, recall, F_1 score, AUC, and accuracy. Precision is a metric that measures the proportion of correct positive predictions (true positives) to total positive predictions (true positives and false positives). Precision gives an idea of how good the model is at identifying the correct samples of the desired class. Recall measures the proportion of correct positive predictions (true positives) to the total samples that are actually positive (true positives and false negatives). Recall gives an idea of how well the model captures all positive samples. F_1 score is a metric that combines information from precision and recall. F_1 score is the harmonic mean of precision and recall and can provide an overall picture of the model's performance in detecting a particular class. The metrics used are defined as follows:

$$P = \frac{TP}{TP + FP} \quad (4)$$

$$R = TPR = \frac{TP}{TP + FN} \quad (5)$$

$$F_1 = \frac{2PR}{P + R} \quad (6)$$

$$Acc = \frac{TP + TN}{TP + FN + FP + TN} \quad (7)$$

$$FPR = \frac{FP}{TN + FP} \quad (8)$$

In this work, P denotes precision and R denotes recall. Accordingly, we compute the F_1 -score as $F_1 = \frac{2 \times P \times R}{P + R}$, which is the harmonic mean of precision and recall (Eq. (6)). Furthermore, we present the ROC (Receiver Operating Characteristics) curve's AUC (Area Under the Curve). AUC is a metric used to evaluate the ability of a binary classification model to distinguish between classes. It measures the performance across all possible classification thresholds, providing a robust assessment of the model's predictive power. The ROC curve depicts the trade-off between the true positive rate (TPR or recall) and false positive rate (FPR) across various classification thresholds. Then, AUC is determined by calculating the area under the ROC curve.

4. Experimental results

All experiments were carried out on the server of the Intelligent Systems laboratory of the Faculty of Computer Science, Brawijaya University, with Intel® Xeon® Gold 6230R CPU @ 2.10 GHz, NVIDIA Quadro RTX 8000 48 GB GDDR6 GPU, and 250 GB RAM.

We evaluated the performance impact of weighted loss on the fine-tuned InceptionV3, ResNet50, DenseNet121, and MobileNetV2 architectures by using nine datasets extracted from SMDG-19 in Table

2. The training was performed with a 70:15:15 split ratio, an Adam optimizer with a learning rate of 0.001, and 200 epochs. To measure the impact of weighted loss, we calculate the percentage of performance change, defined as follows:

$$\text{Change (\%)} = \frac{(v_w - v_b)}{v_b} \times 100\% \quad (9)$$

where v_w is the performance value obtained using weighted loss and v_b is the baseline performance using binary cross-entropy loss. This equation expresses the relative improvement (or decline), where the percentage change is calculated with respect to the baseline value. All reported percentage changes in this study follow this relative formulation.

We also performed a performance comparison with the commonly used focal loss for imbalanced data. Specifically, we implemented and tested the focal loss as defined by Lin et al. [41].

When weighted loss is applied to the InceptionV3 model, precision drops by 13% while recall increases by 42%. The F_1 score has dropped by 2%, while the AUC has improved by 0.1%. Accuracy, on the other hand, decreases by 13%. The result is shown in Table 3, notably, InceptionV3 yielded zero recall and F_1 -score on the G1020 dataset before applying weighted loss, indicating a complete failure to detect any glaucoma cases. This extreme result highlights the critical impact of class imbalance, where the model becomes overly biased toward the majority class. After applying weighted loss, the model's performance improved substantially, demonstrating the effectiveness of this strategy in recovering sensitivity on highly imbalanced datasets.

Weighted ResNet50 model shows a significant 17% decrease in precision but a remarkable 61% improvement in recall. Both the F_1 score (4%) and the AUC (2.6%) have improved. However, accuracy experiences a significant 9% drop. The result is shown in Table 4.

Weighted DenseNet121 has a minor 0.8% decrease in precision but a significant 43% increase in recall. Both the F_1 score (13%) and the AUC (6%) have improved significantly. Accuracy experiences a slight setback, decreasing by 1.5%. The result is shown in Table 5.

Weighted MobileNetV2 model shows optimal changes, with a 2.5% increase in precision and a 13.5% increase in recall. The F_1 score rises 6.5%, whereas the AUC rises 3.3%. However, accuracy is reduced by 1.3%. The result is shown in Table 6.

Across the five metrics used, the focal loss generally increases precision and prediction accuracy. However, this increase is not accompanied by a corresponding improvement in the F_1 score; in fact, the F_1 score is negatively impacted. Conversely, our weighted loss approach successfully improves the F_1 score while also increasing the AUC. Nonetheless, these improvements come with limitations, as accuracy and precision are compromised.

Table 3
Weighted InceptionV3 result with performance change percentage; bold indicates the performance improvement.

No	Split Data	IR	InceptionV3 BCE					InceptionV3 Focal					InceptionV3 Weighted				
			P	R	F1	AUC	Acc	P	R	F1	AUC	Acc	P	R	F1	AUC	Acc
1	SMDG-19	1.59	0.7900	0.8800	0.8300	0.8693	0.8674	0.9000 13.92%	0.8000 -9.09%	0.8500 2.41%	0.8721 0.32%	0.8896 2.56%	0.8400 6.33%	0.8400 -4.55%	0.8400 1.20%	0.8718 0.28%	0.8788 1.31%
2	FIVES	1.67	0.6700	0.6700	0.6700	0.7619	0.8000	0.6900 2.99%	0.5000 -25.37%	0.5800 -13.43%	0.7024 -7.81%	0.7833 -2.09%	0.5600 -16.42%	0.5600 -16.42%	0.5600 -16.42%	0.6825 -10.42%	0.7333 -8.33%
3	G1020	2.45	0.0000	0.0000	0.0000	0.5000	0.7255	0.0000 NaN	0.0000 NaN	0.0000 NaN	0.5000 0.00%	0.7255 0.00%	0.2700 NaN	0.8600 NaN	0.4100 NaN	0.4826 -3.48%	0.3137 -56.76%
4	BEH	2.71	0.7100	0.8000	0.7500	0.8242	0.8333	1.0000 40.85%	0.4000 -50.00%	0.5700 -24.00%	0.7000 -15.07%	0.8125 -2.50%	0.7100 0.00%	0.8000 0.00%	0.7500 0.00%	0.8242 0.00%	0.8333 0.00%
5	ORIGA	2.87	0.7400	0.5200	0.6100	0.7240	0.8163	1.0000 35.14%	0.0400 -92.31%	0.0700 -88.52%	0.5185 -28.38%	0.7347 -10.00%	0.3400 -54.05%	0.9300 78.85%	0.5000 -18.03%	0.6294 -13.07%	0.4898 -40.00%
6	sjchoi86-HRF	2.97	0.9400	0.7500	0.8300	0.8628	0.9016	0.9200 -2.13%	0.6000 -20.00%	0.7300 -12.05%	0.7878 -8.69%	0.8525 -5.45%	0.8200 -12.77%	0.7000 -6.67%	0.8134 -8.43%	0.8525 -5.73%	0.8525 -5.45%
7	PAPILA	3.83	0.8800	0.5400	0.6700	0.7592	0.8889	0.6200 -29.55%	0.6200 14.81%	0.6200 -7.46%	0.7577 -0.20%	0.8412 -5.36%	0.9100 3.41%	0.7700 42.59%	0.8300 23.88%	0.8746 15.20%	0.9365 5.36%
8	REFUGE	9.00	0.7000	1.0000	0.8200	0.9867	0.9750	0.8300 18.57%	0.7100 -29.00%	0.7700 -6.10%	0.8527 -13.58%	0.9750 0.00%	1.0000 42.86%	0.7100 -29.00%	0.8300 1.22%	0.8571 -13.14%	0.9833 0.85%
9	OIA ODIR	14.26	0.8600	0.2300	0.3600	0.6116	0.9355	0.6200 -27.91%	0.1900 -17.39%	0.2900 -19.44%	0.5895 -3.62%	0.9265 -0.96%	0.2300 -73.26%	0.8500 269.57%	0.3600 0.00%	0.8032 31.33%	0.7646 -18.27%
Average		4.59	0.6989	0.5989	0.6156	0.7666	0.8604	0.7311	0.4289	0.4978	0.6978	0.8379	0.6311	0.7800	0.6489	0.7599	0.7540
Average performance change (%)								6.49%	-28.54%	-21.08%	-8.56%	-2.64%	-12.99%	41.80%	-2.07%	0.11%	-13.48%

Table 4
Weighted ResNet50 result with performance change percentage; bold indicates the performance improvement.

No	Split Data	IR	ResNet50 BCE					ResNet50 Focal					ResNet50 Weighted				
			P	R	F1	AUC	Acc	P	R	F1	AUC	Acc	P	R	F1	AUC	Acc
1	SMDG-19	1.59	0.8100	0.8400	0.8200	0.8599	0.8636	0.9400 16.05%	0.7500 -10.71%	0.8400 2.44%	0.8615 0.18%	0.8874 2.76%	0.7800 -3.70%	0.8700 3.57%	0.8200 0.00%	0.8602 0.03%	0.8582 -0.63%
2	FIVES	1.67	0.5600	0.5000	0.5300	0.6666	0.7333	0.7300 30.36%	0.6100 22.00%	0.6700 26.42%	0.7579 13.70%	0.8167 11.37%	0.5700 1.79%	0.4400 -12.00%	0.5000 -5.66%	0.6508 -2.37%	0.7333 0.00%
3	G1020	2.45	0.3900	0.2100	0.2800	0.5441	0.6928	0.0000 -100.00%	0.0000 -100.00%	0.0000 -100.00%	0.4910 -9.76%	0.7124 2.83%	0.2700 -30.77%	1.0000 376.19%	0.4300 53.57%	0.5000 -8.11%	0.2745 -60.38%
4	BEH	2.71	0.6200	0.7700	0.6900	0.7773	0.7813	0.8900 43.55%	0.5700 -25.97%	0.6900 0.00%	0.7682 -1.17%	0.8438 7.99%	0.6000 -3.23%	0.8000 3.90%	0.6900 0.00%	0.7788 0.19%	0.7708 -1.34%
5	ORIGA	2.87	0.7100	0.4400	0.5500	0.6870	0.7959	0.5000 -29.58%	0.6300 43.18%	0.5600 1.82%	0.6951 1.18%	0.7245 -8.97%	0.5500 -22.54%	0.8500 93.18%	0.6700 21.82%	0.7921 15.30%	0.7653 -3.84%
6	sjchoi86-HRF	2.97	0.8800	0.7500	0.8100	0.8506	0.8852	0.8200 -6.82%	0.7000 -6.67%	0.7600 -6.17%	0.8134 -4.37%	0.8525 -3.69%	0.8900 1.14%	0.8000 6.67%	0.8400 3.70%	0.8756 2.94%	0.9016 1.85%
7	PAPILA	3.83	0.7100	0.7700	0.7400	0.8446	0.8889	1.0000 40.85%	0.4600 -40.26%	0.6300 -14.86%	0.7308 -13.47%	0.8889 0.00%	0.6400 -9.86%	0.6900 -10.39%	0.6700 -9.46%	0.7962 -5.73%	0.8571 -3.58%
8	REFUGE	9.00	1.0000	0.7100	0.8300	0.8571	0.9833	1.0000 0.00%	0.5700 -19.72%	0.7300 -12.05%	0.7857 -8.33%	0.9750 -0.84%	0.7800 -22.00%	1.0000 40.85%	0.8800 6.02%	0.9912 15.65%	0.9833 0.00%
9	OIA ODIR	14.26	0.6400	0.5700	0.6000	0.7692	0.9400	0.7800 21.88%	0.3400 -40.35%	0.4700 -21.67%	0.6657 -13.46%	0.9400 0.00%	0.2500 -60.94%	0.8300 45.61%	0.3900 -35.00%	0.8092 5.20%	0.7916 -15.79%
Average		4.59	0.7022	0.6178	0.6500	0.7618	0.8405	0.3790	0.1581	0.2283	0.3452	0.4309	0.5922	0.8089	0.6544	0.7838	0.7706
Average performance change (%)								1.81%	-19.83%	-13.79%	-3.95%	1.27%	-16.68%	60.84%	3.89%	2.57%	-9.30%

4.1. Weighted loss impact on model architectures

Across all four models, the weighted versions consistently show improvements in recall and AUC, indicating enhanced ability to capture positive instances and better overall performance in distinguishing between positive and negative instances. However, there is a trade-off observed with a decrease in precision and accuracy, highlighting the importance of considering specific application requirements when choosing the most suitable model.

Table 7 includes both percentage changes and corresponding absolute values for clarity. Our findings, as summarized in Table 7, show that the use of weighted loss functions resulted in consistent improvements in recall and AUC across all four models. For example,

DenseNet121 improved recall from 64.3% to 91.9%, and F1 score from 64.9% to 73.5%, while maintaining stable precision (from 67.9% to 67.4%).

Weighted DenseNet121 stands out with the best improvements in F1 score and AUC, showing a balanced improvement in precision and recall. On the other hand, MobileNetV2 demonstrates the greatest improvement in precision and accuracy, reflecting its capability for enhanced positive instance recognition and overall correctness.

Overall, the weighted loss function significantly improved the model's ability to capture glaucoma cases, as reflected in increased recall and AUC. These improvements are crucial in clinical screening contexts, where minimizing missed glaucoma cases is vital for early diagnosis and intervention.

Table 5
Weighted DenseNet121 result with performance change percentage; bold indicates the performance improvement.

No	Split Data	IR	DenseNet121 BCE					DenseNet121 Focal					DenseNet121 Weighted				
			P	R	F1	AUC	Acc	P	R	F1	AUC	Acc	P	R	F1	AUC	Acc
1	SMDG-19	1.59	0.8000	0.8600	0.8300	0.8642	0.8642	0.9300 16.25%	0.7600	0.8300	0.8606	0.8853	0.7800	0.8400	0.8100	0.8487	0.8501
								-11.63%	0.00%	0.00%	-0.42%	2.44%	-2.50%	-2.33%	-2.41%	-1.79%	-1.63%
2	FIVES	1.67	0.5600	0.5600	0.5600	0.6825	0.7333	0.6700 19.64%	0.5600	0.6100	0.7183	0.7833 6.82%	0.5600	0.7800 39.29%	0.6500 16.07%	0.7579 11.05%	0.7500
								0.00%	0.00%	8.93%	5.24%	0.00%	0.00%	0.00%	0.00%	0.00%	2.28%
3	G1020	2.45	0.2300	0.1400	0.1800	0.4813	0.6340	0.0000 -100.00%	0.0000 -100.00%	0.0000 -100.00%	0.5000 3.89%	0.7255 14.43%	0.3400 47.83%	0.4800 242.86%	0.4000 122.22%	0.5669 17.79%	0.6078
								-14.52%	-9.93%	-6.33%	5.33%	0.7300	0.8000	0.7600	0.8318	0.8438	0.00%
4	BEH	2.71	0.6000	0.9000	0.7200	0.8136	0.7813	0.7800 30.00%	0.6000	0.6800	0.7621	0.8229	0.7300	0.8000	0.7600 5.56%	0.8318 2.24%	0.8438 8.00%
								-20.00%	0.00%	-11.11%	-3.68%	-5.36%	21.67%	-11.11%	5.56%	2.24%	8.00%
5	ORIGA	2.87	0.5600	0.7000	0.6200	0.7462	0.7653	0.5000 -10.71%	0.5600	0.5300	0.6721	0.7245	0.5500	0.7800 11.43%	0.6500 4.84%	0.7692 3.08%	0.7653
								-20.00%	0.00%	-14.52%	-9.93%	-5.33%	-1.79%	0.6500	0.7400	0.8006	0.00%
6	sjchoi86-HRF	2.97	0.8900	0.8000	0.8400	0.8756	0.9016	0.9100 2.25%	0.5000	0.6500	0.7378	0.8197	0.8700	0.6500	0.7400	0.8006	0.8525
								-37.50%	-22.62%	-15.74%	-9.09%	-2.25%	-18.75%	-11.90%	-8.57%	-5.45%	-5.45%
7	PAPILA	3.83	0.7500	0.6900	0.7200	0.8162	0.8889	0.6000 -20.00%	0.6900	0.6400	0.7862	0.8413	0.9100 21.33%	0.7700 11.59%	0.8300 15.28%	0.8746 7.16%	0.9365 5.35%
								0.00%	0.00%	-11.11%	-3.68%	-5.36%	21.33%	11.59%	15.28%	7.16%	5.35%
8	REFUGE	9.00	1.0000	0.7100	0.8300	0.8571	0.9833	1.0000 0.00%	1.0000 40.85%	1.0000 20.48%	1.0000 16.67%	1.0000 1.70%	0.7800 -22.00%	1.0000 40.85%	0.8800 6.02%	0.9912 15.65%	0.9833 0.00%
								0.5300	0.3000	0.3900	0.6395	0.9235	0.2200	0.7400 72.09%	0.3400 -37.04%	0.7580 6.81%	0.7766 -17.51%
9	OIA ODIR	14.26	0.7200	0.4300	0.5400	0.7097	0.9415	-26.39%	-30.23%	-27.78%	-9.89%	-1.91%	-69.44%	0.7400 72.09%	0.3400 -37.04%	0.7580 6.81%	0.7766 -17.51%
								0.5300	0.3000	0.3900	0.6395	0.9235	0.2200	0.7400 72.09%	0.3400 -37.04%	0.7580 6.81%	0.7766 -17.51%
	Average	4.59	0.6789	0.6433	0.6489	0.7607	0.8326	0.2795	0.1695	0.2116	0.3597	0.4231	0.6378	0.7600	0.6733	0.7999	0.8184
	Average performance change (%)							-9.88%	-21.32%	-16.91%	-2.24%	1.00%	-0.79%	42.88%	13.18%	5.93%	-1.45%

Table 6
Weighted MobileNetV2 result with performance change percentage; bold indicates the performance improvement.

No	Split Data	IR	MobileNetV2 BCE					MobileNetV2 Focal					MobileNetV2 Weighted				
			P	R	F1	AUC	Acc	P	R	F1	AUC	Acc	P	R	F1	AUC	Acc
1	SMDG-19	1.59	0.8300	0.8500	0.8400	0.8706	0.8755	0.9200 10.84%	0.7600	0.8300	0.8604	0.8847 1.05%	0.8200	0.8600 1.18%	0.8400	0.8738 0.37%	0.8772
2	FIVES	1.67	0.5600	0.5600	0.5600	0.6825	0.7333	0.6700 19.64%	0.3300	0.4400	0.6310	0.7500	0.6500	0.8300 48.21%	0.7300 30.36%	0.8214 20.35%	0.8167 11.37%
3	G1020	2.45	0.3700	0.2400	0.2900	0.5425	0.6797	0.5300 43.24%	0.1900	0.2800	0.5637 3.91%	0.7320	0.3500	0.3500 45.83%	0.3500 20.69%	0.5524	0.6405
4	BEH	2.71	0.7100	0.7300	0.7200	0.7985	0.8229	0.8800 23.94%	0.5000	0.6400	0.7348	0.8229	0.6500	0.8700 19.18%	0.7400 2.78%	0.8273 3.61%	0.8125
5	ORIGA	2.87	0.5000	0.5200	0.5100	0.6607	0.7245	1.0000 100.00%	0.2200	0.3600	0.6111	0.7857 8.45%	0.5600	0.7000 34.62%	0.6200 21.57%	0.7462 12.94%	0.7653
6	sjchoi86-HRF	2.97	0.9300	0.6500	0.7600	0.8128	0.8689	0.9000 -3.23%	0.4500	0.6000	0.7128	0.8033	1.0000 7.53%	0.5500	0.7100	0.7750	0.8525
7	PAPILA	3.83	0.5500	0.8500	0.6700	0.8331	0.8254	1.0000 81.82%	0.3800	0.5600	0.6923	0.8730	0.7500	0.6900	0.7200 7.46%	0.8162	0.8889 7.69%
8	REFUGE	9.00	0.8800	1.0000	0.9300	0.9956	0.9917	1.0000 13.64%	0.1400	0.2500	0.5714	0.9500	0.8600	0.8600	0.8600	0.9241	0.9833
9	OIA ODIR	14.26	0.6500	0.5300	0.5800	0.7519	0.9400	0.6200 -4.62%	0.5500	0.5800	0.7589	0.9370	0.4400	0.6400 20.75%	0.5200 -10.34%	0.7857 4.50%	0.9070
Average		4.59	0.6644	0.6589	0.6511	0.7720	0.8291	0.8356	0.3911	0.5044	0.6818	0.8376	0.6756	0.7056	0.6767	0.7913	0.8382
Average performance change (%)								31.70%	-36.66%	-19.69%	-10.13%	1.46%	2.48%	13.51%	6.49%	3.30%	1.29%

Table 7
Average performance change on models across nine datasets, bold indicates the best improvement, and underlined bold indicates the second-best improvement on each metric.

No	Model	Precision (%)	Recall (%)	F1 Score (%)	AUC (%)	Acc (%)
1	InceptionV3	-12.99 (from 69.9% to 63.1%)	41.80 (from 59.9% to 78.0%)	-2.07 (from 61.6% to 64.9%)	0.11 (from 76.7% to 76.0%)	-13.48 (from 86.0% to 75.4%)
2	ResNet50	-16.68 (from 70.2% to 59.2%)	60.84 (from 61.8% to 80.9%)	3.89 (from 65.0% to 65.4%)	2.57 (from 76.2% to 78.4%)	-9.30 (from 84.1% to 77.1%)
3	DenseNet121	<u>-0.79</u> (from 67.9% to 63.8%)	42.88 (from 64.3% to 76.0%)	13.18 (from 64.9% to 67.3%)	5.93 (from 76.1% to 80.0%)	-1.45 (from 83.3% to 81.8%)
4	MobileNetV2	2.48 (from 66.4% to 67.6%)	13.51 (from 65.9% to 70.6%)	6.49 (from 65.1% to 67.7%)	3.30 (from 77.2% to 79.1%)	1.29 (from 82.9% to 83.8%)

Among the four models, DenseNet121 with weighted loss showed the best balance across all performance metrics, particularly in terms of F1 score and AUC. This makes it the most clinically viable candidate for glaucoma screening applications.

Table 8

Detailed performance measure of DenseNet121 classifier using 5-fold cross validation.

DenseNet121												
No	Dataset	IR	Precision (%)		Recall (%)		F1 Score (%)		AUC (%)		Acc (%)	
			Mean	95% CI	Mean	95% CI	Mean	95% CI	Mean	95% CI	Mean	95% CI
1	SMDG-19	1.59	94.51	[90.70, 98.32]	90.19	[82.76, 97.62]	92.25	[86.50, 98.00]	93.50	[88.81, 98.19]	94.26	[90.23, 98.29]
2	FIVES	1.67	82.09	[70.44, 93.74]	86.45	[78.40, 94.50]	83.95	[75.01, 92.89]	87.56	[80.80, 94.31]	87.75	[81.16, 94.34]
3	G1020	2.45	52.45	[33.96, 70.95]	47.66	[20.24, 75.08]	48.66	[24.37, 72.95]	66.94	[53.42, 80.45]	75.39	[67.86, 82.93]
4	BEH	2.71	79.10	[72.81, 85.39]	79.94	[75.07, 84.82]	79.20	[76.07, 82.33]	85.98	[84.35, 87.61]	88.80	[87.17, 90.44]
5	ORIGA	2.87	81.95	[68.86, 95.04]	72.71	[51.77, 93.65]	75.25	[59.57, 90.93]	83.18	[71.74, 94.62]	88.00	[80.06, 95.94]
6	sjchoi86-HRF	2.97	76.67	[64.34, 89.00]	83.84	[75.73, 91.95]	78.93	[73.62, 84.24]	87.73	[84.48, 90.97]	89.03	[86.55, 91.51]
7	PAPILA	3.83	92.61	[88.40, 96.81]	83.87	[76.48, 91.27]	87.91	[82.46, 93.35]	91.02	[86.91, 95.13]	95.00	[92.40, 97.60]
8	REFUGE	9.00	94.48	[87.84, 101.11]	95.48	[89.74, 101.21]	94.84	[89.64, 100.05]	97.46	[94.46, 100.47]	99.00	[98.08, 99.92]
9	OIA-ODIR	14.26	77.28	[65.59, 88.97]	60.56	[40.01, 81.10]	66.34	[49.97, 82.71]	79.71	[69.39, 90.03]	96.24	[94.53, 97.96]
Weighted DenseNet121												
1	SMDG-19	1.59	90.91	[85.17, 96.66]	93.46	[89.16, 97.77]	92.13	[87.37, 96.88]	93.74	[89.92, 97.55]	93.79	[90.01, 97.57]
2	FIVES	1.67	79.46	[66.37, 92.55]	87.49	[74.06, 100.92]	82.49	[71.42, 93.56]	86.71	[78.45, 94.97]	86.50	[78.43, 94.57]
3	G1020	2.45	42.71	[24.13, 61.29]	51.06	[36.62, 65.50]	45.43	[29.21, 61.65]	59.18	[46.07, 72.30]	62.45	[47.65, 77.25]
4	BEH	2.71	76.80	[70.39, 83.20]	85.63	[78.50, 92.75]	80.77	[75.13, 86.41]	87.93	[83.98, 91.88]	89.12	[86.03, 92.21]
5	ORIGA	2.87	73.54	[60.64, 86.44]	82.39	[65.29, 99.48]	76.03	[64.66, 87.40]	85.32	[77.03, 93.61]	86.31	[78.98, 93.64]
6	sjchoi86-HRF	2.97	88.50	[79.04, 97.97]	80.37	[64.15, 96.59]	82.80	[74.92, 90.67]	88.62	[81.16, 96.07]	92.03	[88.91, 95.15]
7	PAPILA	3.83	86.13	[77.49, 94.77]	91.18	[85.05, 97.30]	88.30	[82.37, 94.23]	93.26	[89.43, 97.09]	94.53	[91.02, 98.03]
8	REFUGE	9.00	86.37	[72.69, 100.04]	93.59	[88.99, 98.19]	88.94	[81.28, 96.61]	95.90	[94.03, 97.77]	97.75	[96.27, 99.23]
9	OIA-ODIR	14.26	61.29	[47.09, 75.49]	78.44	[69.40, 87.48]	68.29	[56.60, 79.99]	87.36	[82.13, 92.60]	95.07	[92.76, 97.38]

Table 9

Average performance change on datasets across four model architectures, bold indicates the best improvement, and underlined bold indicates the second-best improvement on each metric.

No	Dataset	IR	Precision (%)	Recall (%)	F1 Score (%)	AUC (%)	Acc (%)
1	SMDG-19	1.59	-0.27	-0.53	-0.30	-0.28	-0.19
2	FIVES	1.67	0.36	14.77	6.09	4.65	1.33
3	G1020	2.45	3.88	221.63	65.49	4.08	-17.00
4	BEH	2.71	2.50	2.99	2.08	1.51	1.35
5	ORIGA	2.87	-16.59	54.52	7.55	4.56	-9.55
6	sjchoi86-HRF	2.97	-1.59	-8.53	-5.80	-4.00	-2.73
7	PAPILA	3.83	12.81	6.24	9.29	3.65	3.71
8	REFUGE	9.00	-0.85	9.67	1.44	2.74	0.00
9	OIA-ODIR	14.26	-58.99	102.01	-20.60	11.96	-13.77
Average		4.59	-6.53	44.75	7.25	3.21	-4.10

4.2. Results with cross validation

We further performed a 5-fold cross validation on the datasets using our best performing model architecture, DenseNet121. The datasets were randomly divided into five equal portions. In each training session, one part was reserved for testing while the remaining four were used for training. This process was repeated five times, with each part being used once for testing. The average was taken over the five training sessions. The glaucoma class precision, recall, F1 scores, AUC, and accuracy are tabulated in Table 8.

The table demonstrates that using weighted loss improved classification performance in terms of recall, F1 score, and AUC. However, a slight decrease in precision and accuracy was observed because the model focused more on the minority class (glaucoma). Most improvements were seen in datasets with a higher imbalance ratio (IR). Additionally, the improvement is supported by the narrower 95% confidence interval observed with the weighted loss approach.

4.3. Weighted loss impact on datasets

Since each individual dataset has its own characteristics, we analyzed the average of each dataset's performance change across four model architectures. Then the result is shown in Table 9. Using this table, we plot all datasets against each metric to find a trend in the dataset's imbalance ratio towards performance changes.

Fig. 4 shows at a glance that the application of weighted loss across all datasets does not have a significant trend towards changes in precision in the glaucoma class. Broadly speaking, it can be seen

that 5 out of 9 datasets, or 55.56% of the total datasets, experienced a decrease in precision performance, which accounts for a total average change in precision due to the application of the weighted loss of -6.53%. In addition, the OIA-ODIR dataset is the dataset with the largest decrease in precision. This is related to OIA-ODIR's imbalance ratio of 14.26, which indicates that this dataset is very unbalanced. OIA-ODIR is also the dataset with the highest imbalance ratio, far from the average imbalance ratio of the datasets used in this research.

Furthermore, Fig. 5 also does not show any significant trend in the recall change for the glaucoma class across all datasets. However, 7 out of 9 datasets, or 77.78% of the total datasets, experienced an increase in recall performance which accounts for a total average recall change due to the application of the weighted loss technique of 44.75%, including OIA-ODIR which experienced the largest increase in precision performance previously. The most affected dataset in terms of recall performance is G1020 which has a recall increase of 221.63%. G1020 itself has an IR of 2.45 which does not show a significant imbalance. However, out of the four architectures used in this research, G1020 is the lowest performing dataset, whether using weighted loss or not, even G1020 performances without weighted loss on the InceptionV3 architecture have zero values that are successfully improved when using weighted loss.

Finally, Fig. 6 also does not show any significant trend in the change of F1 score of the glaucoma class across all datasets. However, 6 out of 9 datasets, or 66.67% of the total datasets, experienced an improvement in F1 score performance which accounts for a total average F1 score change due to the application of the weighted loss technique of 7.25%.

This research shows that the application of weighted loss on glaucoma detection performance successfully boosted the model's attention

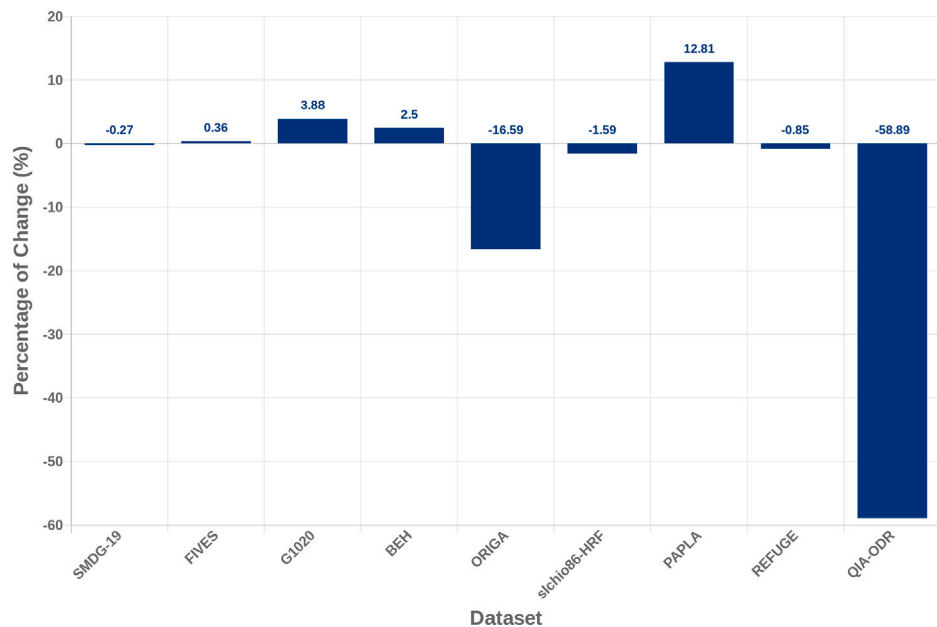


Fig. 4. Percentage change in precision after applying weighted loss across nine datasets. Positive values indicate improved precision; negative values indicate a drop.

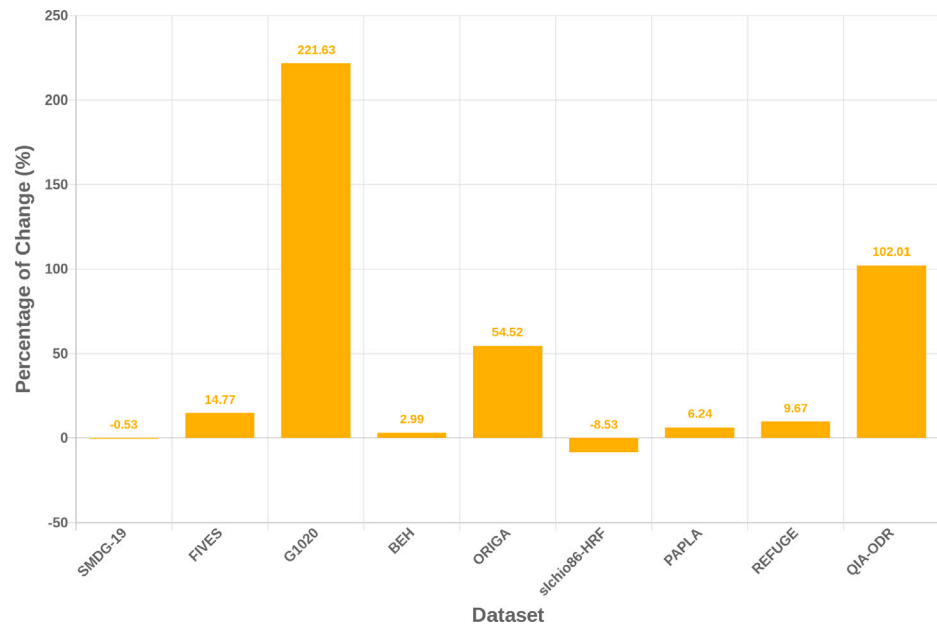


Fig. 5. Percentage change in recall after applying weighted loss across nine datasets. Positive values indicate improved sensitivity to glaucoma detection.

in predicting glaucoma class, resulting in an increase in the frequency of glaucoma predictions. The four models examined saw an average relative recall improvement of 44.75% as the number of glaucoma class predictions increased. It should be emphasized, however, that as the number of glaucoma class predictions increases, the models become more likely to detect non-glaucoma classes inaccurately. This is demonstrated by the glaucoma class's average drop in precision of -6.53%. This suggests that the model can improve glaucoma class detection performance by 44.75% by sacrificing 6.53% precision. This is supported by a 7.25% increase in F1 score, indicating that the trade-off between precision and recall does not degrade the level of balance between the two metrics. Furthermore, no significant trend exists between the influence of weighted loss on the imbalance ratio across the nine datasets evaluated.

Fig. 7 shows at a glance that the application of weighted loss across the datasets has a slightly positive trend of increasing the change in AUC. Broadly speaking, it can be seen that 7 out of 9 datasets, or 77.78% of the total number of datasets, experienced an increase in AUC, which accounts for the total average AUC change due to the application of weighted loss increasing by 3.21%. In addition, OIA-ODIR is the dataset with the largest average AUC increase of 11.96%. This further shows that OIA-ODIR, with an IR of 14.26, is the most unbalanced and most affected dataset by the use of weighted loss.

Furthermore, Fig. 8 shows no trend in accuracy change across the datasets. However, 5 out of 9 datasets, or 55.56% of the total number of datasets, experienced a decrease in accuracy which contributed to a total average change in accuracy due to the application of weighted loss decreasing by -4.10%. The most affected dataset's accuracy is G1020

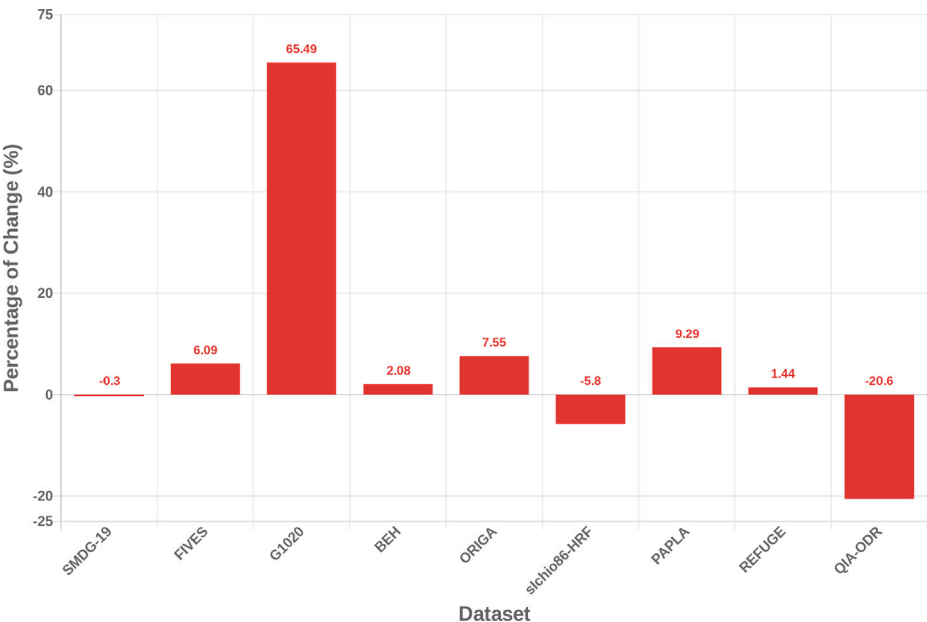


Fig. 6. Percentage change in F1-score after applying weighted loss across glaucoma datasets. The F1-score balances precision and recall, making it suitable for evaluating performance on imbalanced datasets.

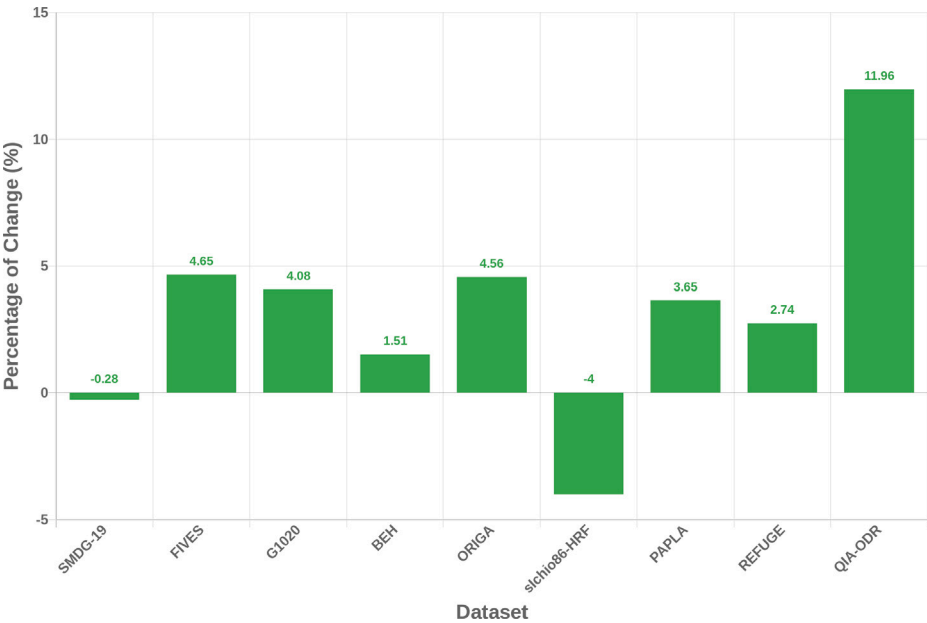


Fig. 7. Percentage change in AUC (Area Under the Curve) after applying weighted loss across multiple datasets. AUC measures the model's ability to distinguish between classes across decision thresholds. Positive values reflect improved discriminative capability after reweighting.

with a decrease of -17.00% , followed by OIA-ODIR with a decrease of -13.77% .

These figures reveal that applying the weighted loss technique to the overall model performance on the above 9 datasets can provide an average increase in AUC of 3.21% and a decrease in accuracy of -4.10% . Both results indicate that the application of weighted loss on the unbalanced dataset successfully improved the model's attention to the glaucoma class samples. However, the decrease in accuracy indicates that the accuracy obtained without using weighted loss is a “blind” accuracy which could be a percentage of the majority class so that with the application of weighted loss, the accuracy becomes more realistic that the model is not performing as well as the “blind” accuracy value.

This research managed to outperform most of the performance of previous state-of-the-art methods through the application of weighted loss. However, the precision, AUC, and accuracy performance of ORIGA are still lower than those of previous researches because this research only improves the recall and F1 score performance of the ORIGA dataset, while the performance of most other datasets is improved in this research. This comparison can be seen in [Table 10](#).

One of the main challenges in comparing various datasets is the image characteristics of a dataset. Bajwa et al. [43] applied Principal Component Analysis (PCA) to obtain the two most significant principal components and visualized them in 2D on the ORIGA and G1020 datasets. He found that glaucoma images and healthy images can be well separated in the ORIGA dataset. However, in the G1020 dataset,

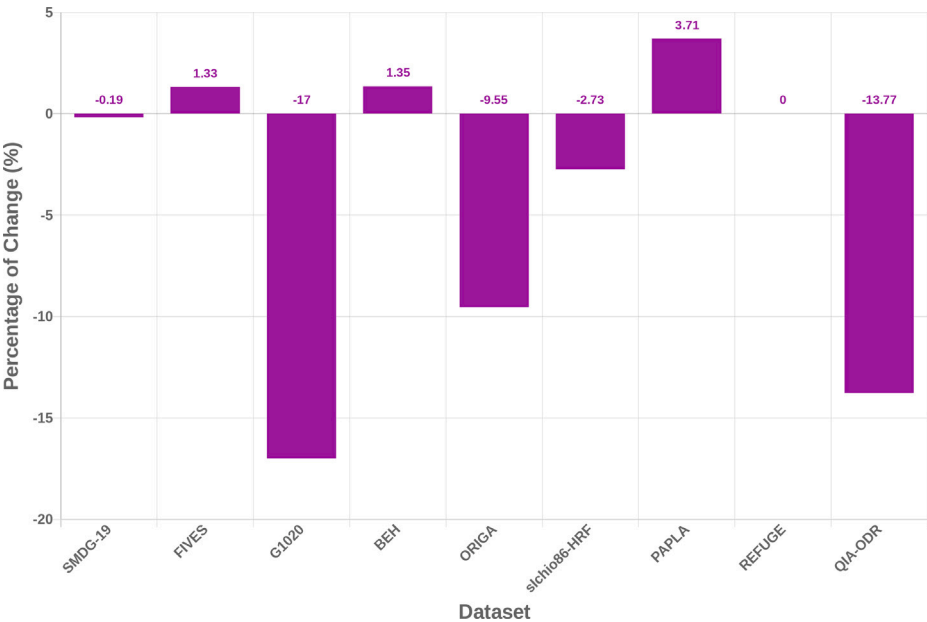


Fig. 8. Percentage change in accuracy after applying weighted loss across nine glaucoma-related datasets. Positive bars indicate improvement, while negative values indicate a drop in overall classification accuracy.

Table 10
Performance metrics comparison for glaucoma detection; bold indicates the best performance for each dataset in each metric.

Author	Model	Dataset	Precision (%)	Recall (%)	F1 Score (%)	AUC (%)	Acc (%)
[42]	Deep CNN	ORIGA	66.00	44.00	52.00	87.40	–
		G1020	10.00	33.00	15.00	–	–
[43]	InceptionV3	ORIGA	69.00	56.00	62.00	85.00	–
		G1020	29.00	19.00	22.00	49.00	–
[44]	Xception	sjchoi86-HRF	–	70.33	–	77.39	70.82
[45]	Ensemble	ORIGA	–	–	–	83.59	88.86
		REFUGE	–	–	–	95.10	95.59
[46]	DenseNet121	PAPILA	–	–	–	80.00	–
This research	InceptionV3	ORIGA	34.00	93.00	50.00	62.94	48.98
		G1020	27.00	86.00	41.00	48.26	31.37
		sjchoi86-HRF	82.00	70.00	76.00	81.43	85.25
		REFUGE	100.00	71.00	83.00	85.71	98.33
		PAPILA	91.00	77.00	83.00	87.46	93.65
		ORIGA	55.00	85.00	67.00	79.21	76.53
	ResNet50	G1020	27.00	100.00	43.00	50.00	27.45
		sjchoi86-HRF	89.00	80.00	84.00	87.56	90.16
		REFUGE	78.00	100.00	88.00	99.12	98.33
		PAPILA	64.00	69.00	67.00	79.62	85.71
	DenseNet121	ORIGA	55.00	78.00	65.00	76.92	76.53
		G1020	34.00	48.00	56.69	60.78	60.78
		sjchoi86-HRF	87.00	65.00	74.00	80.06	85.25
		REFUGE	78.00	100.00	88.00	99.12	98.33
		PAPILA	91.00	77.00	83.00	87.46	93.65
	MobileNetV2	ORIGA	56.00	70.00	62.00	74.62	76.53
		G1020	35.00	35.00	35.00	55.24	64.05
		sjchoi86-HRF	100.00	55.00	71.00	77.50	85.25
		REFUGE	86.00	86.00	86.00	92.41	98.33
		PAPILA	75.00	69.00	72.00	81.62	88.89

the two classes have a very large overlap which makes it difficult to separate the two classes properly. Therefore, the effect of weighted loss also varies depending on the characteristics of the dataset itself.

Table 11 presents the averaged specificity, negative predictive value (NPV), and false positive rate (FPR) for each model-loss configuration of ORIGA dataset. These metrics provide critical insights into the real-world clinical applicability of each approach. High specificity is essential in glaucoma screening to avoid unnecessary anxiety and

resource overuse caused by false positives. In this context, InceptionV3-BCE and InceptionV3-Weighted demonstrated near-perfect specificity (1.000 and 0.968), implying excellent reliability in ruling out non-glaucoma patients.

Similarly, NPV reflects the probability that a negative result truly indicates the absence of glaucoma. Models such as DenseNet121-Focal (NPV = 0.832) and MobileNetV2-Focal (NPV = 0.812) offer high confidence in negative predictions, which is clinically valuable to reassure patients and reduce follow-up burden.

Table 11
Averaged Specificity, NPV, and FPR across G1020 dataset for each model and loss function.

Model	Loss	Specificity (avg)	NPV (avg)	FPR (avg)
InceptionV3	BCE	1.000	0.727	0.000
	Focal	0.218	0.772	0.782
	Weighted	0.968	0.778	0.032
ResNet50	BCE	0.888	0.782	0.112
	Focal	0.376	0.467	0.624
	Weighted	0.902	0.779	0.098
DenseNet121	BCE	0.900	0.773	0.100
	Focal	0.704	0.832	0.296
	Weighted	0.809	0.793	0.191
MobileNetV2	BCE	0.966	0.763	0.034
	Focal	0.773	0.812	0.227
	Weighted	0.826	0.780	0.174

However, Focal loss consistently increases the false positive rate across all models up to 78% in InceptionV3 and 62% in ResNet50. Such high FPR can lead to overdiagnosis and strain on healthcare systems. This trade-off underscores the importance of balancing sensitivity with specificity.

In contrast, weighted loss functions consistently provided a favorable balance. For instance, MobileNetV2-Weighted achieved high specificity (0.826), acceptable NPV (0.780), and a controlled FPR (0.174), suggesting that weighting schemes improve both sensitivity and model restraint.

Clinically, these findings highlight that optimal model selection should not rely solely on recall or accuracy. Specificity, NPV, and FPR directly affect clinical outcomes, affecting whether patients are unnecessarily referred or falsely reassured. Weighted loss emerges as a practical and medically sound compromise for deep learning-based glaucoma screening systems.

4.4. Grad-CAM visualization

The Grad-CAM result in Fig. 9 shows the performance of three model architectures in predicting unweighted and weighted glaucoma fundus image samples. This figure shows the lack of performance of the model in noticing important parts of the fundus image without the application of weighted loss. However, the model with weighted loss applied is quite capable of paying attention to important parts of the fundus image, namely around the optic cup of the fundus image. This means that the model succeeds in paying more attention when encountering glaucoma images.

In particular, for ResNet50, DenseNet121, and MobileNetV2 architectures, the visualizations reveal that the unweighted models often generate scattered or misaligned attention maps, focusing on irrelevant peripheral areas such as the lower or central fundus regions.

After applying the weighted loss function, there is a consistent improvement across all architectures, with clearer and more concentrated attention around the optic disc and neuroretinal rim. This is especially evident in the DenseNet121 and MobileNetV2 visualizations, where the heatmaps shift from non-diagnostic zones to clinically meaningful locations associated with glaucomatous changes. These findings indicate that incorporating weighted loss enhances the model’s ability to focus on diagnostic features, reduces distraction from irrelevant regions, and improves interpretability, which is crucial for deployment in clinical decision support systems.

To investigate the interpretability of the trained deep learning model in classifying glaucoma, we applied Grad-CAM on a series of test images. The results were analyzed based on two groups: images predicted as glaucoma (Fig. 10(a)) and images predicted as healthy (Fig. 10(b)).

In the glaucoma-predicted images (Figs. 10(a)), the attention heatmaps focused predominantly on the optic disc area, particularly around

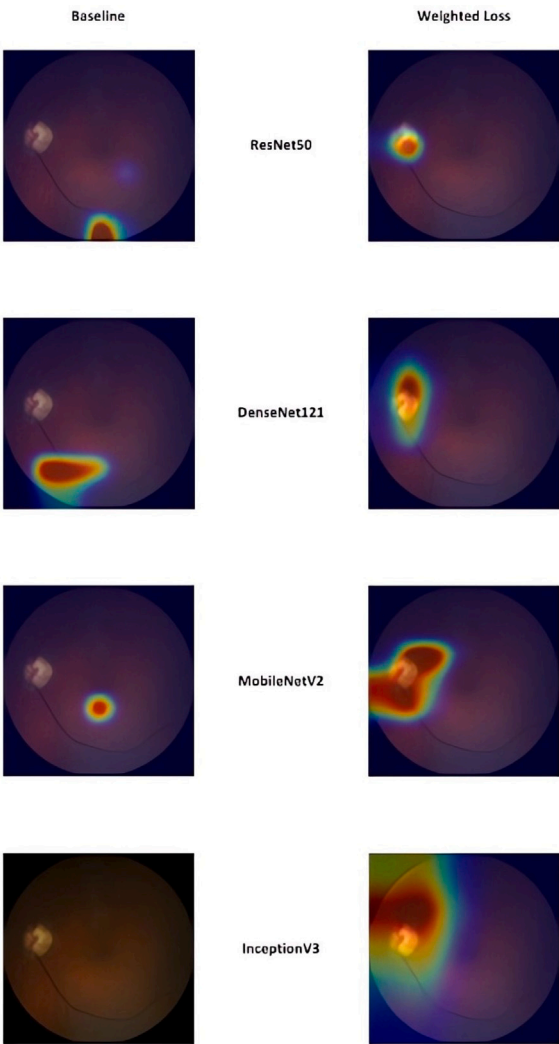


Fig. 9. Grad-CAM visualization on a random fundus image of without and with Weighted Loss.

the neuroretinal rim and the cup region. These regions are clinically relevant, as glaucomatous damage is commonly characterized by optic disc cupping, rim thinning, and loss of the nerve fiber layer. High intensity activation, represented by red and yellow regions, suggests that the model successfully localizes structural features indicative of glaucoma.

In contrast, the healthy-predicted images (Fig. 10(b)) exhibited broader and more peripheral attention distributions. In particular, the optic disc received comparatively lower attention in most cases, and some models instead highlighted the macula or the retinal background. This may reflect the model’s assessment of the absence of pathological cues, strengthening its decision to use a healthy label.

Interestingly, some variations were observed even within each category. For example, while the second row of Fig. 10(a) still focused on the optical disc, the spatial spread of the attention was more diffuse than in the first row of Fig. 10(a), indicating the uncertainty of the model or the variation in the data. Conversely, healthy images such as Fig. 10(b) showed highly scattered heatmaps, consistent with the model recognizing a lack of localized abnormality.

The Grad-CAM results support the model’s ability to focus on clinically significant areas during classification. The localization of attention in glaucoma cases aligns with medical diagnostic criteria, while the dispersed patterns in healthy cases suggest confidence in normalcy.

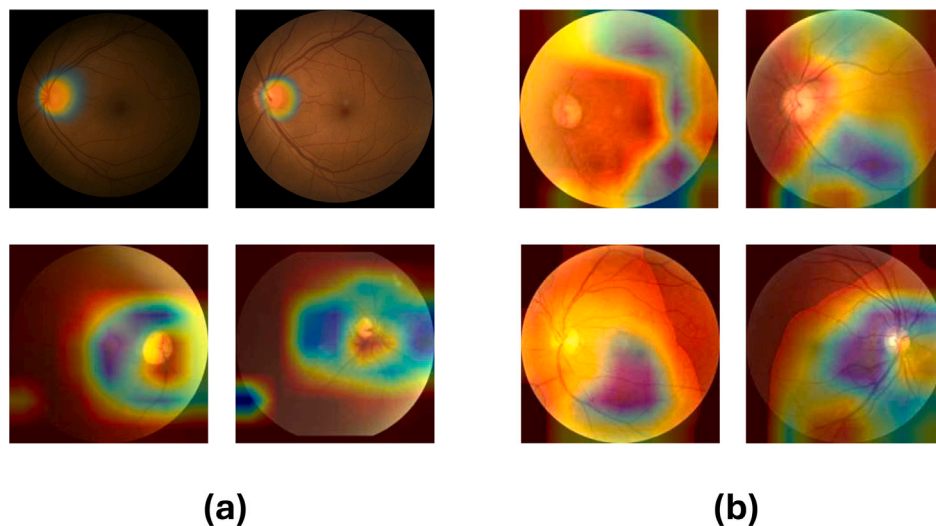


Fig. 10. Grad-CAM visualization results for glaucoma and healthy fundus images. (a) Grad-CAM heatmaps for fundus images classified as glaucoma. The activation maps demonstrate strong attention around the optic disc and neuroretinal rim, which aligns with clinically relevant regions affected by glaucomatous damage. (b) Grad-CAM heatmaps for fundus images classified as healthy. The attention is more diffusely distributed, often away from the optic disc, indicating the absence of focal pathological features. These results suggest the model effectively distinguishes pathological and non-pathological patterns by focusing on relevant anatomical structures in glaucoma classification.

These visualizations contribute to the transparency of the model and improve trustworthiness in clinical applications.

4.5. Inference time processing

DenseNet-121 model achieves an inference latency of approximately 3 to 8 ms per image on the Quadro RTX 8000, translating to a throughput of 125 to 300 images per second, depending on optimization. When processing images in batches, the GPU efficiently leverages parallelism. For instance, while a single image may require 7–8 ms, a batch of 10 images processed in only 50–60 ms due to amortized overhead and parallel computation. This high throughput allows real-time analysis of dozens of fundus images per second, enabling large-scale screening without perceptible latency.

In clinical settings, such as primary care or integrated fundus camera systems, this efficiency permits near-instantaneous inference. The DenseNet model can flag suspect glaucoma cases immediately after image acquisition, supporting timely triage and referral. Importantly, the latency introduced by the CNN is minimal compared to other factors like image capture or data transfer. However, it is crucial to note that without hardware acceleration, such as on general-purpose CPUs or mobile-class ARM cores, inference times can increase dramatically (e.g., up to 3–4 s per image), underscoring the importance of GPU or AI accelerator deployment. Encouragingly, recent mobile hardware (e.g., Qualcomm AI cores) have demonstrated DenseNet-121 inference latencies as low as 2–8 ms, making real-time screening feasible even on edge devices. Nevertheless, final clinical decisions should continue to rely on comprehensive, multimodal evaluations.

5. Discussion

The consistent underperformance of baseline models trained using standard BCE loss across multiple CNN architectures and datasets highlights a critical limitation in handling class-imbalanced medical imaging data. In this study, baseline models frequently exhibited suppressed recall and poor F1-scores, particularly on datasets with pronounced imbalance ratios. For example, the InceptionV3 and ResNet50 models completely did not detect any positive samples (recall = 0%) on the G1020 dataset, illustrating a scenario in which the classifier was completely dominated by the majority class (non-glaucoma). This phenomenon reflects the inherent bias of BCE, which assigns equal

importance to all classes and thus favors the majority class during optimization.

Weighted loss functions offer a principled solution to this challenge by rebalancing the loss contribution from each class, thereby encouraging the model to learn discriminative features even from underrepresented glaucoma cases. Empirical results demonstrated substantial gains in recall across all tested architectures, with ResNet50 achieving a relative improvement of +60.84% and DenseNet121 improving by +42.88%. In parallel, F1-score and AUC also improved for all models, suggesting that weighted loss enhances both sensitivity and the model's ability to separate positive and negative classes. The improvements were particularly notable on datasets such as OIA-ODIR (IR = 14.26), where baseline performance was suboptimal and recall increased markedly after weighting.

Nevertheless, the increased sensitivity introduced by the weighted loss came at the expense of decreased precision and accuracy. The average precision declined by −6.53%, and accuracy dropped by −4.10%, indicating a higher rate of false positives. While this trade-off may be unfavorable in general-purpose classification tasks, it is acceptable — and even desirable — in medical screening contexts, where failing to detect true cases (false negatives) carries higher clinical risk than over-diagnosis. These findings underscore the importance of task-specific model selection and loss function design in medical applications.

However, while statistically significant, the improvements in recall (e.g., +44.75%) and F1-score (+7.25%) may not always lead to clinical decision changes. To contextualize the observed performance changes with weighted loss, Table 12 summarizes the clinical implications of key metric changes and proposes potential clinical actions in response to different model outputs. This emphasizes the role of the model in helping early detection and prioritization rather than autonomous decision making. The model is best interpreted as a screening tool to identify possible glaucoma cases for further evaluation. Precision losses (6.53%) imply higher false positive rates, but can be managed through clinical confirmation. Despite this, the improved F1 score (+7.25%) reflects a better balance between sensitivity and specificity, strengthening the utility of the model as a triage aid. The slight drop in accuracy (−4.10%) highlights the trade-off when prioritizing minority-class detection in imbalanced datasets, underscoring that accuracy alone may not fully capture screening performance. Lastly, the increase in AUC (+3.21%) suggests enhanced discriminatory power across thresholds, making the model adaptable for use in diverse clinical settings through

Table 12
Practical implications of metric changes and corresponding clinical actions.

Metric Change with Weighted Loss	Practical Implication (Clinical Context)	Potential Clinician Action
↑ Recall (+44.75%)	More glaucomatous cases correctly identified; fewer missed diagnoses (i.e., reduced false negatives). Particularly beneficial in early screening.	Refer patients flagged by the model for further testing (e.g., OCT, visual field); prioritize early detection in at-risk populations.
↓ Precision (−6.53%)	Increased false positives may lead to unnecessary follow-ups or anxiety. Higher burden on referral systems.	Confirm flagged cases with clinical judgment; use as triage tool, not stand-alone diagnosis.
↑ F1-Score (+7.25%)	Improved harmonic balance between precision and recall; suggests enhanced reliability in screening settings.	Use model outputs as supportive evidence when prioritizing patients for further glaucoma assessment.
↓ Accuracy (−4.10%)	Slight decline in overall correctness; due to increased detection of minority class (glaucoma), potentially at the cost of majority class accuracy.	Accuracy should not be solely relied upon in imbalanced settings; prioritize sensitivity for screening.
↑ AUC (+3.21%)	Better ability to distinguish between glaucomatous and normal discs over various thresholds; useful in threshold tuning.	Consider integrating model into screening systems with adjustable thresholds for different population risk levels.

threshold tuning. Thus, the system supports, but does not replace, expert diagnosis.

Beyond quantitative improvements, qualitative insights obtained through Grad-CAM visualizations further validated the role of weighted loss in enhancing model interpretability. Baseline models frequently focused on irrelevant fundus regions, producing diffuse and clinically uninformative activation maps. In contrast, models trained with weighted loss showed consistent focus on anatomically relevant regions such as the optic cup and neuroretinal rim—key structures in glaucoma assessment. This focused attention aligns with clinical expectations and supports the model’s trustworthiness as a decision-support tool.

Importantly, the impact of weighted loss was not uniform across all datasets. While datasets with extreme imbalance ratios (e.g., OIA-ODIR and REFUGE) experienced the most pronounced gains, improvements were also observed in moderately imbalanced datasets (e.g., FIVES and BEH), suggesting the generalizability of this approach. The cross-architecture consistency of performance gains further supports the robustness of weighted loss, with DenseNet121 demonstrating the most balanced trade-off among recall, precision, F1-score, and AUC.

Taken together, these findings confirm that the baseline failure to detect minority cases stems from a fundamental mismatch between standard loss functions and imbalanced medical datasets. The use of weighted loss not only improves quantitative performance on underrepresented classes but also enhances qualitative interpretability, making it a compelling choice for glaucoma classification and other high-stakes diagnostic applications.

Beyond performance metrics, the proposed model also holds potential clinical utility as a screening tool, particularly in resource-limited settings. Its high sensitivity and NPV suggest that it could be effectively used to rule out glaucoma in patients who do not require immediate specialist referral. This is especially valuable in primary care or community health scenarios, where ophthalmologists and advanced imaging technologies like (OCT) may not be readily available.

Because the model relies solely on fundus images, which are relatively inexpensive and widely available, it could support early detection efforts in low- and middle-income regions. By enabling automated triage, the system could help prioritize patients for further examination, reduce diagnostic delays, and improve access to timely glaucoma care in underserved populations.

6. Limitations

This study has several limitations that should be acknowledged. First, although the proposed method was validated using the comprehensive SMDG-19 dataset comprising images from 19 public sources,

all evaluations were conducted retrospectively. As such, the generalizability of the model to prospective clinical environments remains to be confirmed. Differences in image quality, acquisition devices, and population characteristics could impact real-world performance.

Furthermore, the ground-truth labels across the 19 datasets were determined by different groups of clinicians or trained graders, primarily based on fundus photograph evaluation, often supplemented by clinical data. This introduces potential inconsistency in diagnostic standards, as optic disc assessment is known to suffer from inter-observer variability even among glaucoma specialists. The absence of uniform, objective confirmatory tests (e.g., visual fields or OCT means that some “glaucoma” labels may reflect subjective or early findings. Hence, relying solely on human-labeled fundus images as gold standards can result in label noise and potential bias.

Second, the classification task was restricted to a binary outcome (glaucoma vs. non-glaucoma), which does not capture the full clinical spectrum. Early-stage glaucoma, glaucoma suspects, and other optic neuropathies were not differentiated. Future models should incorporate more granular labels or staging schemes to improve clinical relevance.

Third, the model is trained exclusively on monocular 2D fundus photographs, which inherently lack stereoscopic or depth information crucial for evaluating the three-dimensional morphology of the optic nerve head (ONH). Clinical assessment of glaucomatous damage, particularly features such as cup depth, neuroretinal rim thinning, and cup-to-disc ratio, requires spatial depth cues that are absent in flattened 2D projections. Without this information, both human experts and deep learning models may fail to accurately detect subtle structural changes, especially in early or borderline glaucoma. In contrast, stereo fundus imaging allows clinicians to visualize the optic disc in 3D, improving both sensitivity and interobserver agreement. Studies have shown that stereo viewing yields higher diagnostic accuracy and better agreement with glaucoma specialists compared to monoscopic images [47,48]. Moreover, the monocular nature of the fundus images used omits stereo depth perception, which could have improved detection of optic cup depth and structural contours. Future work should consider stereo imaging or OCT data to enhance performance.

Furthermore, depth-resolved modalities such as OCT provide objective, quantitative measurements of the ONH and retinal nerve fiber layer (RNFL), including cup volume and rim area, which have been validated as reliable markers of glaucomatous damage [49]. Integrating OCT with fundus photography has been shown to significantly improve diagnostic performance. For example, Islam et al. demonstrated that a multimodal deep learning model combining fundus and OCT data achieved an AUC of 0.95, outperforming models trained on fundus-only (AUC 0.89) or OCT-only data (AUC 0.87) [50]. These findings

underscore a key limitation of monocular 2D imaging and highlight the potential of incorporating stereo or volumetric data into AI pipelines to better mimic clinical decision-making and enhance model robustness.

Fourth, our input data lacks functional information such as visual field (VF) data, which is central to glaucoma diagnosis and progression monitoring. Visual field testing reflects the functional integrity of the visual system and captures aspects of neural damage not visible on structural imaging. Numerous studies have emphasized that functional loss, especially in early glaucoma, can precede or exist independently of the structural damage observed in fundus photography or OCT [51]. In particular, VF loss correlates strongly with the patient's quality of life, impacting daily activities such as reading, mobility, and driving. Medeiros et al. demonstrated a significant relationship between the rate of VF loss and declines in vision-related quality of life, highlighting the clinical and patient-reported relevance of this modality [52].

The omission of this modality limits the clinical utility and interpretability of our deep learning model to assess the severity of the disease. Deep learning systems trained on structural images alone may underestimate or overlook functional impairment, leading to suboptimal decision-making in both screening and staging [53].

Moreover, we did not incorporate ocular vascular parameters such as peripapillary capillary perfusion, despite growing evidence linking vascular dysfunction to glaucomatous damage. Optical Coherence Tomography Angiography (OCTA) studies have shown that glaucomatous eyes often exhibit reduced peripapillary vessel density and deep-layer microvascular dropout, both of which are associated with progressive VF loss and structural thinning [54,55]. These vascular abnormalities may precede detectable changes on conventional imaging, providing an earlier signal of glaucomatous damage.

Unfortunately, fundus photographs, while convenient and accessible, do not provide perfusion or hemodynamic information. This absence can lead to misclassification, particularly in cases where the optic disc appears structurally normal, yet there is underlying vascular compromise or perfusion instability [56]. Thus, future work should aim to integrate multimodal data, including VF and OCTA, to enhance the sensitivity and generalizability of glaucoma detection models.

Fifth, the use of a weighted loss function improved recall and F1-score, but these improvements, while statistically significant, may not always translate into meaningful clinical decision changes for individual patients. For example, a small increase in predicted probability from 90% to 92% is unlikely to affect treatment. The benefit of the weighted model is more prominent at the population level, particularly in screening settings where increased sensitivity reduces the number of missed cases. However, precision drops slightly, increasing false positives, which may burden downstream confirmatory assessments.

Finally, we did not conduct subgroup analysis based on patient demographics (e.g., age, ethnicity) or image acquisition devices. Such factors can influence model performance and bias, particularly in globally deployed systems. Future work should explore fairness-aware modeling and external validation across diverse clinical environments to ensure equity and robustness.

7. Conclusion

This study evaluated four deep learning architectures on nine glaucoma-related datasets to assess the effectiveness of weighted loss functions in improving the detection of underrepresented classes. We compared the performance of weighted loss and focal loss against the standard binary cross-entropy loss. While focal loss may enhance precision in certain scenarios, it often leads to a reduction in recall and F1-score, which is less desirable for early-stage disease detection where missing true cases can have critical clinical implications.

In contrast, the application of weighted loss consistently improved the model's sensitivity to glaucoma cases by reducing false negatives, even though it introduced a slight increase in false positives. This

trade-off was beneficial overall, resulting in more balanced and robust classification performance. Moreover, the use of weighted loss enhanced the model's ability to distinguish between glaucoma and non-glaucoma images under class imbalance conditions. While a minor reduction in overall accuracy was observed, this is expected as the model reallocates focus toward the minority class.

Interpretability analysis using Grad-CAM further supported these findings. Visualizations showed that models trained with weighted loss attended more consistently to clinically relevant regions, such as the optic disc and neuroretinal rim, whereas baseline models often produced dispersed or less informative attention maps. These results highlight weighted loss as an effective and clinically meaningful strategy for deep learning-based glaucoma detection. It offers improved model responsiveness to early signs of disease while maintaining interpretability and minimizing clinically disruptive errors.

CRedit authorship contribution statement

Devin Jaya Nugraha: Visualization, Software, Investigation, Data curation. **Novanto Yudistira:** Writing – review & editing, Supervision, Methodology, Funding acquisition, Conceptualization. **Agus Wahyu Widodo:** Writing – review & editing, Validation, Project administration.

Ethical statement for computers in biology and medicine

This study utilized publicly available, de-identified human subject data from SMDG-19. The data have been widely used in previous research and was originally collected under approved ethical protocols with informed consent obtained from all participants. As the datasets are publicly accessible and contain no personally identifiable information, the present analysis was conducted in accordance with relevant ethical guidelines and did not require additional ethical approval from our institutional review board.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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